

AI Employee Productivity Platform – Solution Architecture & NFR

Document Control

Document ID: EAI-ARCH-002 | Version: 1.7 | Owner: Enterprise AI Architecture | Status: Architecture Review

Overview

The platform combines a web workspace, FastAPI services, LangGraph agent orchestration, specialized agents, Azure AI Search, PostgreSQL, Microsoft Graph integrations and Azure AI Foundry model endpoints.

Agents

Email Agent extracts decisions/actions. Chat Agent analyzes collaboration messages. Document Agent retrieves content. Functional Agent builds business views. Technical Agent builds architecture views. Synthesis Agent combines evidence.

Model Policy

Only organization-approved Azure foundation models are permitted. Agents request capabilities such as reasoning, context capacity and latency; model routing selects an approved endpoint.

RAG

Content is ingested, parsed, normalized, chunked, enriched with metadata and indexed. Retrieval combines semantic and vector techniques and preserves source references.

Availability

Production target availability is 99.9%, subject to approved platform and dependency SLAs.

Performance

Simple retrieval queries should meet the agreed interactive response target. Workflows implement bounded timeouts and retries.

Scalability

Agent services should be stateless where practical. Ingestion and enrichment use asynchronous processing and support horizontal scaling.

Observability

Capture request traces, retrieval details, model endpoint, tool calls, latency, errors and evaluation metrics.

Governance

Prompts, models, retrieval configuration and agent workflows must be version controlled and evaluated before promotion.